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{ "model_choice": { "champion": { "name": "google/gemma-3-27b-it", "rationale": "Early-fusion
DeepStack directly projects images into the main residual stream, bypassing cross-attention
bottlenecks. Its native interleaved local/global attention mathematically isolates intra-page
layouts (local sliding window) while routing inter-page context (global attention) across a
massive 128K context window without  $O(N^2)$  memory explosion." },
"easier_surgery_open_us_vlm": { "name": "microsoft/Phi-4-multimodal-instruct", "why_not":
"Standard dense transformer with naive early-fusion makes RoPE patching trivial. However,
 $O(N^2)$  dense global attention guarantees catastrophic VRAM explosion and spatial attention
dilution when scaling beyond 4 pages. It lacks the sparse/interleaved routing needed for deep
document multi-vector extraction." }, "high_capacity_weak_surgery_us_vlm": { "name":
"meta-llama/llama-3.2-90B-Vision-Instruct", "why_not": "Fatal architectural flaw for ColPali:
'Cross-Attention Quarantine'. Vision tokens are locked in the encoder cache and only activated
via text-query cross-attention. ColPali indexing is text-less and offline; therefore, visual patches
never enter the main LLM residual stream to interact across pages without a massive,
unfeasible pretraining rewrite." } }, "architecture_patch_plan": { "exact_files_to_modify": [
"src/transformers/models/gemma3/modeling_gemma3.py",
"src/transformers/models/gemma3/configuration_gemma3.py" ], "minimal_patch_set": [ "1.
Subclass Gemma3RotaryEmbedding into Volumetric3DRoPE to process pos_ids of shape
[Batch, 3, SeqLen] (Z=Page, Y=Row, X=Col), splitting inverse frequencies across axes.", "2.
Override Gemma3Model._update_causal_mask to yield an OmniDirectionalMask (all 1s)
exclusively for visual tokens when is_document_indexing=True.", "3. Append ColPaliHead via
nn.Linear(config.hidden_size, 128, bias=False) tied to the final layer's hidden states.", "4. Apply
F.avg_pool1d(kernel_size=4, stride=4) on the sequence dimension before the projection to
compress multi-vector index bloat by 75%." ], "exact_expected_tensor_shapes": { "input_pixels":
"[Batch, Num_Pages, Channels, H, W] (e.g., [1, 4, 3, 896, 896])", "early_fusion_embeddings":
"[Batch, Total_Seq_Len, D_model] (e.g., [1, 16384, 4608])", "3d_position_ids": "[Batch, 3,
Total_Seq_Len]", "q_k_states_pre_rope": "[Batch, Num_Heads, Total_Seq_Len, Head_Dim]",
"terminal_hidden_states": "[Batch, Total_Seq_Len, 4608]", "pooled_multi_vectors": "[Batch,
Total_Seq_Len / 4, 128] (L2 Normalized)" }, "positional_encoding_entry": "Inside
Gemma3Attention.forward(), intercept the sequence right before
apply_rotary_pos_emb(query_states, key_states, cos, sin). The 3D positional coordinates are
evaluated into partitioned cosine/sine matrices and applied via complex multiplication.",
"fallback_if_3d_rope_fails": "Abandon intra-attention 3D-RoPE to prevent pre-trained weight
collapse. Revert to standard 2D-RoPE, and inject a LearnedPageSegmentEmbedding
(nn.Embedding(MAX_PAGES, D_model)). Add this vector directly to the token embeddings at
Layer 0 based on their <page_break> index." }, "prototype_14_day_plan": [ { "phase": "Days
1-3: Scaffold & Surgery", "goal": "Establish monkey-patching and verify tensor flow.", "tasks": [
"Implement Volumetric3DRoPE, OmniDirectionalMask, and pooling head in
modeling_gemma3.py.", "Run offline indexing forward passes with dummy tensors representing
1 to 10 pages.", "Verify OOM safety (VRAM scaling) and assert exact tensor dimensions." ] }, {
"phase": "Days 4-7: Contrastive Manifold Anchoring", "goal": "Teach the base model dense
visual retrieval.", "tasks": [ "Freeze base LLM backbone. Train ONLY the 128-D ColPali
projection head and 3D-RoPE embeddings.", "Use single-page and dual-page synthetic ViDoRe
data with standard InfoNCE MaxSim loss.", "Validate single-page zero-shot retrieval beats
random baseline to prove gradient flow." ] }, { "phase": "Days 8-11: RL Judge Integration",
"goal": "Push inter-page reasoning via reward shaping.", "tasks": [ "Unfreeze top 6 global
attention blocks (LoRA r=64).", "Implement CPROG-guided GRPO training loop on the query
side.", "Train on synthetic datasets requiring explicit cross-page <think> trajectories." ] }, {

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"phase": "Days 12-14: Ablation & Deployment", "goal": "Benchmark and finalize Phase 1 prototype.", "tasks": ["Run degradation benchmarks (1 vs 2 vs 4 pages).", "Evaluate multi-page cross-attention failure taxonomy.", "Export weights and test int8 Binary Quantization in Vespa/Qdrant."]], "judge_strategy": { "assessment": "Vanilla GRPO relies on sparse scalar outcome rewards (e.g., nDCG@5) and its PPO-style trust-region limits exploration, causing policy collapse on continuous 128-D retrieval spaces. DISPO decouples up/down clipping of importance weights, preventing this collapse during heavy exploration of multi-page pathways. CPROG provides strict structural checks on the reasoning trace. We require a Hybrid: Contrastive Pretraining to build the vector manifold, followed by DISPO-optimized RL, guided by a CPROG programmatic reward function.", "reward_decomposition": { "retrieval_hit_rank_reward": "+0.50 * continuous nDCG@5 (calculated strictly via MaxSim dot-product against ground-truth document vectors).", "cross_page_consistency_reward": "+0.30 (via CPROG) if the <think> trace programmatic parser verifies synthesis of anchor entities from at least two distinct page_idx spatial zones.", "hallucination_penalty": "-0.30 if the generated <think> trace references a spatial layout coordinate or numeric entity mathematically absent from the retrieved vectors.", "formatting_traceability_reward": "+0.10 for strict XML adherence to the <think>, <page_idx>, and <retrieve> operational tags." } }, "ablation_matrix": { "cross_page_chart_table_query_benchmark": { "setup": "1,000 queries explicitly requiring mapping a color-coded Legend on Page 1 to a specific bar in a Chart on Page 2, and multiplying it by a coefficient in a Table on Page 3.", "success_metric": "Top-1 retrieved document chunk must natively encapsulate the exact spatial patches on all required pages via MaxSim. Target: Hit@1 > 85%." }, "retrieval_degradation_curve": { "setup": "Evaluate the exact same single-page query against an index of the document artificially padded into 1-page, 2-page, 4-page, and 10-page chunks.", "success_metric": "The curve must be sub-linear. Less than 4.0 absolute nDCG@5 points lost between 1-page and 10-page contexts." }, "failure_taxonomy": { "wrong_page_z_axis_collapse": "Model retrieves the exact correct layout geometry (e.g., a table) but from the wrong page/year. Indicates 3D-RoPE Z-axis depth failure.", "wrong_section_x_y_smearing": "Target text multi-vectors are blended with adjacent bounding boxes. Indicates the 1D pooling stride (factor=4) is too aggressive.", "single_page_bias_masking_failure": "The model disproportionately weights Page 1 tokens, effectively ignoring Page 4. Signals an Omni-Mask gradient failure during indexing." } }, "go_no_go_gates": [{ "gate": "Gate 1: Tensor & Memory Validation (Day 3)", "criteria": "Forward pass of 10 concatenated high-res pages completes using < 70GB VRAM with Omni-mask intact and outputting exact [Batch, Seq_Len/4, 128] tensors.", "action_if_failed": "Increase Hierarchical Average Pooling factor to 8. Enable Unsloth heavy gradient checkpointing." }, { "gate": "Gate 2: Manifold Warmup (Day 7)", "criteria": "Zero-shot Single-Page nDCG@5 > 82.0 on ViDoRe holdout set (proves the projection head and 3D-RoPE initialization haven't lobotomized pre-trained multimodal alignment).", "action_if_failed": "Revert to 2D-RoPE + Learned Page Embeddings fallback strategy." }, { "gate": "Gate 3: Cross-Page Viability (Day 14)", "criteria": "4-Page Cross-Retrieval benchmark strictly outperforms Qwen2.5-VL ColPali baseline by at least +3.0 absolute MRR.", "action_if_failed": "Halt V1 release. Initiate hyperparameter sweep on CPROG cross-page consistency reward weights and DISPO clipping bounds." }], "top_5_risks_and_mitigations": [{ "risk": "Omnidirectional Masking permanently destroys causal text generation alignment during the RL reasoning phase.", "mitigation": "Strict architectural branching: is_document_indexing=True strictly uses the Omni-mask (no text generation), while is_query=True strictly uses the Causal text mask to preserve instruction-following capabilities." }, { "risk": "3D-RoPE Inverse Frequencies fall severely out-of-distribution compared to Gemma's native base, destroying visual features.", "mitigation": "Initialize the Z (page depth) axis

frequencies to near-zero so it acts as an identity function at Step 0. Initialize X/Y to the exact mapped pre-trained 1D spatial weights." }, { "risk": "Vector Database (Qdrant/Vespa) storage footprint explodes due to concatenating 10+ pages per chunk (O(N) tokens).", "mitigation": "Enforce strict Int8 Binary Quantization (BQ) combined with the 1D Avg Pooling (factor of 4) prior to the ColPali projection head, reducing footprint by >85%." }, { "risk": "RL Judge CPROG reward hacking (model outputs generic 'cross-page' text to farm formatting rewards without actual visual grounding).", "mitigation": "Tie the absolute MaxSim cosine similarity hard-metric as a strict gate multiplier. If true patch-level MaxSim < 0.2, the total RL reward strictly equals 0.0." }, { "risk": "Gemma 3's local sliding window layers fail to cross page boundaries, isolating pages.", "mitigation": "Ensure the document <page_break> tokens mathematically align with the global attention layer cadence (1 in 6 layers) so cross-page semantic routing strictly occurs in global layers." }] }